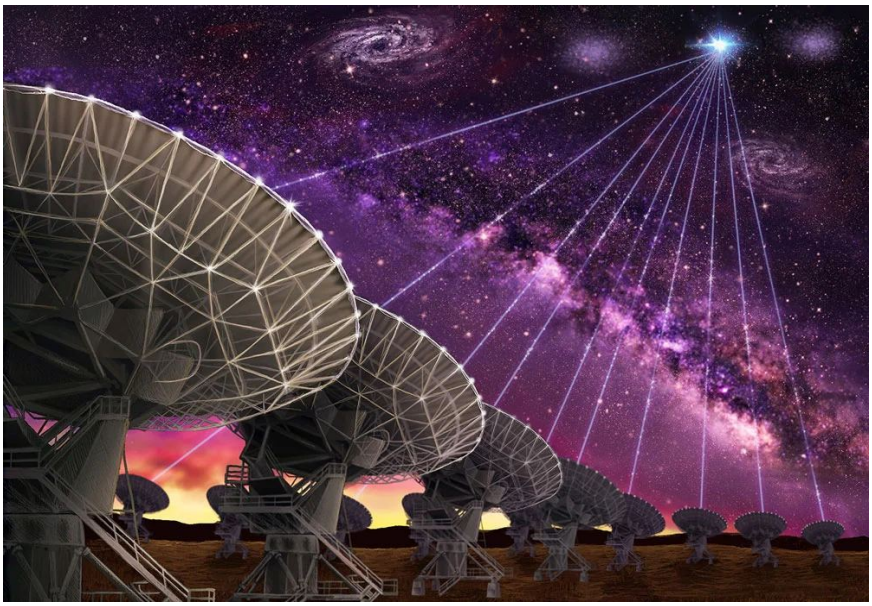
Peering Into the Depths of Our Universe: Seeing to Infinity and Beyond



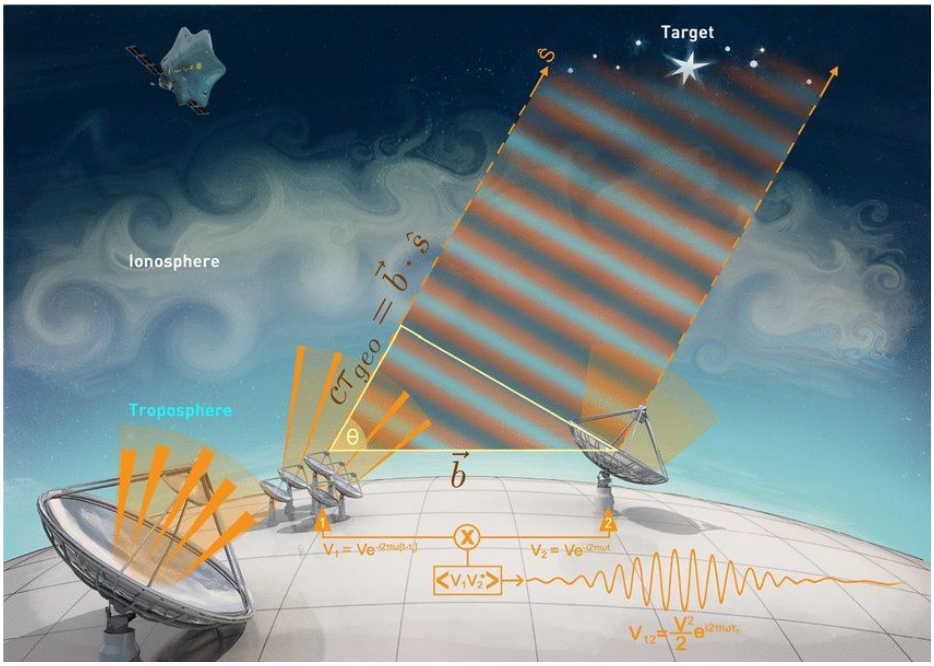
https://www.astronomy.com/science/fast-radio-bursts-now-a-bit-less-mysterious/

In the vastness of our universe exist "whispers". Whispers that are by no means detectable by human ears. They travel through galaxies, star clusters, and space dust, spanning over billions of light years until they reach us. You might ask, "What are these whispers?", "How were we able to detect them?", or how do we even know they exist? Very Long Baseline Interferometry (VLBI) and radio astronomy answer these questions. Acting as our eyes and ears, they can detect what we can't, guiding us through the realm of the unknown and unraveling the secrets of our universe.

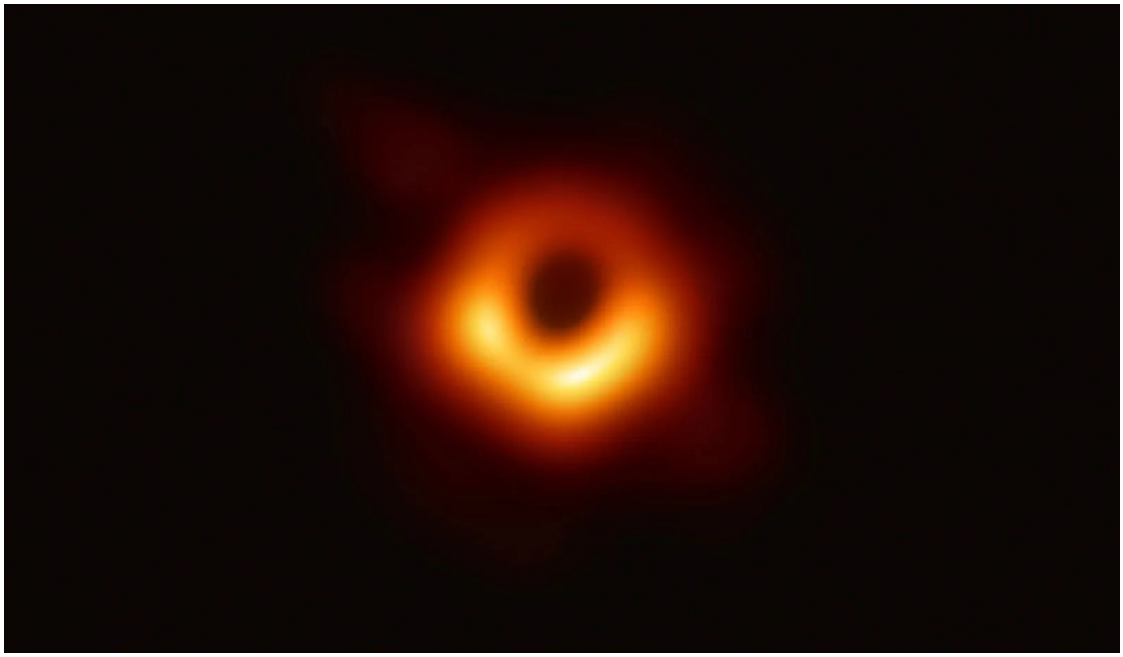
Since the 1960s, VLBI and Radio astronomy have been tools used to measure a variety of phenomena both in and outside our planet. From estimating tectonic shifts to finding out that days are never quite 24 hours. VLBI has historically aided astronomers in discovering the rules that govern our everyday lives and continues to serve as a tool that we use to reveal new information about the cosmos.

Catching Waves

Light emits energy which is referred to as waves. Similar to waves we might see in the ocean, the waves that light emits can vary in size and length. Larger waves have higher energy, and those that are smaller have lower energy. Additionally, the more often a wave occurs, or the more "frequent" it is, then the higher energy it will have, while the less frequent it is, the lower energy it will have; this is what we call the Electromagnetic Spectrum! VLBI primarily deals with radio energy, which has the lowest energy of any type of light on the EM Spectrum. To detect radio energy, astronomers developed a special telescope with a satellite dish on it to intercept the wavelengths and transfer them into images and data.

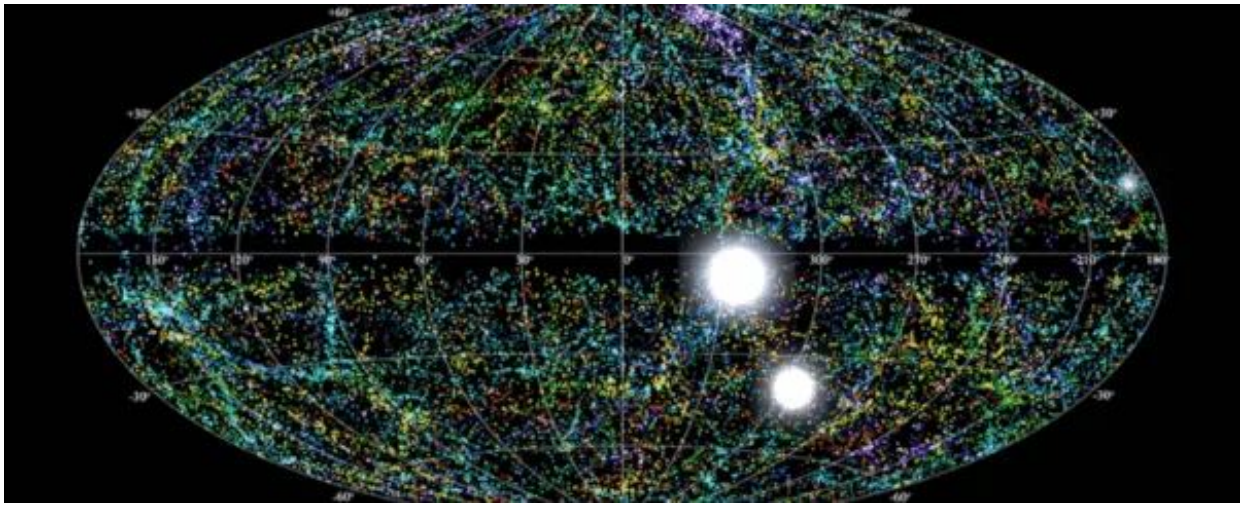


Imagine a calm pool of water, and a rock being thrown into it to create a ripple emitting in all directions. The further the ripples travel, the less visible the wave will be until eventually the wave is nearly undetectable. Now remember, radio energy is already very low, so once its waves reach Earth, they are extremely hard to detect. To resolve this issue, astronomers placed telescopes all across the world, separated by continents, which come together to form one large virtual observatory. With this method, we can get clearer images and better signals of celestial objects, revolutionizing the way we interpret the cosmos, this is VLBI! A recent example of this method is the production of the Event Horizon Telescope (EHT), which was used to capture the first and only image of the black hole at the center of our galaxy, M-87 Star. However, in the wake of astronomical advancement, a new phenomenon emerged; Fast Radio Bursts or what we call *whispers*.



A Blink in Time

Fast Radio Bursts (FRBs) are a rather new phenomenon and field of study within the realm of astrophysics. They are short bursts of radio light that occur in random spots in our universe that shine even brighter than galaxies and last around a one-thousandth of a second. They were accidentally discovered in 2007 by astronomer Duncan Lorimer and his colleagues, searching the night sky for signs of pulsars; large rotating stars emitting radio energy, similar to FRBs. Lorimer and his team were able to differentiate these signals because pulsars typically exert a consistent amount of radio energy, while the burst that he observed was far brighter and more energetic than anything they had ever witnessed. Since their first detection, FRBs have remained a mystery in the realm of astrophysics; astronomers have yet to find a relationship between when and where these galactic whispers will appear.



There are many theories and hypotheses as to what could be causing FRBs, but the widely accepted belief is that the cause of these short bursts is due to "Compact Objects", celestial objects like Black Holes, Neutron Stars, and White Dwarfs. Some believe that the cause of this phenomenon could be a sign of extraterrestrial life or activity. However many theories arise, something prevents any of us from being absolutely certain of the cause of FRBs, and it lies within our reach: The Ionosphere.

Conquering Earths Veil

The Ionosphere is a thick layer of charged particles that absorbs low-energy light, in other words, it is possible to absorb radio light and make the process of FRB detection more difficult by delaying the times at which observatories can receive these signals. The ionosphere is also subject to change, it does not remain consistent which allows astronomers to calculate points in which they can search for FRBs without disruption.



Its elusiveness and mystery are not the only issues though, the ionosphere acts almost as a veil for incoming wavelengths to our satellites which can cause the waves to become distorted, or reach satellites at different times, reducing the accuracy of some VLBI measurements. This is especially so for satellites that focus on the capturing of radio light because it was already hard enough to detect, and the presence of this veil causes a dilemma for modern-day astronomers.

High-Flying Observatories

Astronomers have been attempting to find solutions to the problem that is the ionosphere since the maturation of VLBI. Some astronomers propose to create models that represent the movement of the ionosphere, though this can usually take a long time to make fully functional, and there can be an abundance of error within it, so it is not the most effective method. Some astronomers proposed to use the GPS signal of satellites in space which can continuously monitor the movement of the ionosphere. However, this method can also have its fair share of inaccuracies due to the requirement of precision by VLBI observatories. In the wake of this issue, groundbreaking research has been performed at the MIT Haystack Observatory, where astronomers have proposed to create balloon-born space observatories that will ascend beyond the ionosphere and be operated remotely. This innovative solution can completely revolutionize the world of VLBI as we know it and overcome "Earth's Veil", allowing us to observe our universe more clearly and accurately than ever before.



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